

PROBLEM STATEMENT: Development of an APP for data updation of the Allottees / industries located in SIPCOT Industrial Park on Investment / employment / water / power/ turnover / CSR activities etc.

TITLE - SIPTRACK: SIPCOT Industrial Performance Tracking System

ABSTRACT

SIPCOT (State Industries Promotion Corporation of Tamil Nadu) is a government organization which is responsible for establishing and managing industrial parks across Tamil Nadu to promote industrial growth, investment, and employment. These SIPCOT consist of numerous industries with allotted land and infrastructure facilities such as water supply, power supply, and roads. In order to monitor industrial development and ensure proper utilization of resources, SIPCOT requires industries to regularly submit important operational data such as total investment, employment details, water consumption, power consumption, annual turnover, and Corporate Social Responsibility (CSR) activities. This data is essential for analyzing industrial growth, planning infrastructure improvements, ensuring compliance, and generating reports for administrative and government purposes.

In the present system, the data collection and management of industries are, to some extent, manual and decentralized. The industries submit data in the form of Excel files, emails, hard copies, or simple web forms. The present system has some drawbacks, such as the possibility of data duplication, a lack of provision for real-time updates, difficulties in maintaining a centralized database, increased chances of errors, and a time-consuming process in the generation of reports. In addition, it is difficult for the government to monitor the performance of

industries in different estates and districts because of the absence of a centralized and user-friendly system.

The primary aim of this project is to design and develop a centralized web application for the effective management of industrial information in the SIPCOT estates. The proposed system will develop a secure online platform where the industries can submit and update their operational information online. The system has a multi-level user concept that includes Industry Admin, SIPCOT Estate Admin, and SIPCOT Head Office Admin.

The Industry Admin is the representation of a specific industry operating inside a SIPCOT estate. They can safely log in and update the company information, such as company profile, investment, number of employees, water and power consumption, turnover, and CSR activities. In this manner, the industrial information is always updated in the central database.

The SIPCOT Estate Admin is the representation of the estate authority that takes care of the industries operating inside a specific SIPCOT estate or district. They can safely view, verify, and monitor the information entered by the industries under their control. They can also generate reports regarding the performance of industries at the estate level.

The SIPCOT Head Office Admin is the representation of the central administrative authority and has access to information from all the estates and industries. They can generate overall reports, track the total investment, monitor the employment generation, and analyze the resource consumption from all the SIPCOT estates. This enables the higher authorities to make strategic decisions, develop policies, and plan for the future industrial development.

The proposed system has many advantages over the existing system. The proposed system offers central storage of information, real-time updating, accuracy, and safe role-based access control. The proposed system eliminates

paperwork, reduces human errors, and saves time by automatically producing reports.

The application is developed using modern web technologies based on the client server model, which assists in efficient communication between different parts of the system. The frontend is developed using React.js to provide an interactive and user-friendly interface, which enables users to easily access and use the system features. The backend is developed using Node.js and the Express.js framework, which are responsible for handling the system logic, user authentication, and smooth communication between the frontend and the database. These technologies ensure that the system operates properly and securely. A MySQL database is used to store the data in a structured, secure, and reliable manner. It assists in organizing the data efficiently and enables easy retrieval and management whenever required.

The application is hosted on the Railway cloud platform, which provides reliable and scalable hosting services. The system also provides secure login, proper data management, and REST-based communication, which enables the application to be easily maintained, scalable, and capable of handling large amounts of industrial data.

Keywords: Centralized web application, Role-based access control, Secure login/authentication, Report generation, Client-server architecture, Data management, React.js (frontend framework), Node.js (backend runtime), Express.js (backend framework), MySQL database.